Climate-Aware Stochastic Optimization of Hydropower, Renewable Generation, and Battery Storage for Electricity-System Resilience in Albania

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Research prototype — preprint, not peer reviewed

Manuscript generated from the HYDROFLEX AI experiment pipeline

Abstract

Albania generates almost all of its electricity from hydropower, so rainfall variability translates directly into import dependence. We present HYDROFLEX AI, an open and reproducible linear-programming model that coordinates a physically explicit reservoir water balance, run-of-river hydro, solar, wind, batteries with independent power and energy sizing, and a capacity-limited interconnector. It is solved with the open HiGHS solver and verified against an independent PyPSA formulation (objectives agree to solver precision). The planning layer is a two-stage stochastic programme with a conditional-value-at-risk (CVaR) term, built on hourly representative weeks whose seasonal boundaries are valued at water values from a coarse annual model. The planning results were checked against full-year chronology: representative-week costs are within 1.7–10 MEUR/yr, and the strategies rank in the same order. Inputs are weather-derived capacity factors (NASA POWER), reconstructed inflows and *synthetic* demand and prices.
With a fully available interconnector, a 40% inflow reduction causes no shortage but raises net imports from 0.8 to 3.5 TWh/yr and operating cost by 284 MEUR/yr. Shortages appear only when import availability falls to about half, and they are energy-limited. In that regime batteries pay for themselves through avoided unserved energy, but not through arbitrage alone (13–27% of their cost). Solar PV is a no-regret investment under the base assumptions: its marginal value is about 140 kEUR/MW-yr against an 80 kEUR/MW-yr cost. Because PV dominates, stochastic and risk-averse planning select the same plan as deterministic planning (value of the stochastic solution = 0). When exports are restricted to 300 MW, risk-averse planning adds wind and cuts $\mathrm{CVaR}_{0.9}$ by 74.5 MEUR/yr for 15.1 MEUR/yr of expected cost (full-year chronology). Coarse time steps and pinned week boundaries each distorted the conclusions. The model has not been validated against observed Albanian operations.

1 Introduction

Albania is one of the most hydropower-dependent electricity systems in Europe. Hydropower made up about 95% of installed generating capacity at the end of 2018 (2,096 MW), and the River Drin cascade, made up of Fierza (500 MW), Koman (600 MW) and Vau i Dejës (250 MW), is the backbone of domestic supply [1, 2]. This structure makes annual electricity supply a function of rainfall and snowmelt. Reported net imports are closely correlated with annual precipitation [1], and IRENA's assessment projects that average output from large and small hydropower plants could be roughly 15% and 20% lower by 2050 than in 2010 [1]. Climate variability is therefore not a remote planning consideration for Albania. It is the dominant operational risk.

Three developments make the question of *flexibility* timely. First, utility-scale solar photovoltaics (PV) and batteries have become much cheaper; IRENA reports global weighted-average installed costs of USD 691/kW for PV, USD 1,041/kW for onshore wind and USD 192/kWh for battery storage projects in 2024 [3]. Second, the reservoir fleet is itself a large energy store whose seasonal value depends on how it is coordinated with variable renewables and imports [4]. Third,

regional interconnections let a hydro system import in dry periods and export in wet ones. That exposes the system to regional prices and to the availability of transfer capacity.

This paper asks: *how should a hydropower-dependent electricity system coordinate reservoir operation, renewable generation, battery storage and electricity imports under uncertain rainfall, fluctuating demand and financial constraints?* We address eight supporting questions. They cover the reliability effect of drought, the import-reduction potential of solar, wind and batteries, how storage value changes with drought severity, budget-constrained investment, the performance of plans optimised for average conditions under shocks, the value of stochastic and risk-averse planning, the marginal value of each flexibility resource, and sensitivity to prices, demand, costs and inflows.

Contributions. The work is an independent research prototype with four contributions:

1. An open, fully reproducible linear-programming model of the Albanian system. It couples an hourly energy balance with a physically consistent reservoir water balance (in hm^3, with an explicit energy equivalent), independent battery power and energy sizing, an aggregate interconnector and an unserved-energy penalty. It is solved with the open-source HiGHS solver and verified against an independent PyPSA formulation.

2. A two-stage stochastic investment programme with a Rockafellar–Uryasev CVaR term. Capacities are shared first-stage decisions and dispatch is scenario specific at hourly resolution on representative weeks. Seasonal reservoir coupling is retained through nested annual boundary conditions.

3. A transparent climate-scenario construction that ties drought severities to a reconstructed 2001–2023 inflow index derived from NASA POWER precipitation. Every series is labelled as weather-derived, reconstructed or synthetic.

4. A systematic marginal-value analysis that compares dual-based and finite-difference values of battery energy, battery power, PV, wind, import capacity and reservoir volume, and states explicitly when duals can and cannot be interpreted economically.

The model is embedded in an interactive decision-support application whose numbers come exclusively from executed optimisation runs. We stress at the outset what the work is not. It is not an operational model, it has not been validated against observed Albanian grid operations, and its demand and price inputs are synthetic (Section 10).

2 Literature review

Hydropower-dependent systems and climate risk. Hydro-dominated systems face supply risk that is driven by water availability rather than by generating capacity. At the global scale, van Vliet et al. [5] showed that a large share of hydropower and thermoelectric capacity is vulnerable to reduced streamflow and warmer water under climate change. They also argued that adaptation options such as efficiency gains and fuel switching matter. For Albania, IRENA's readiness assessment highlights both the dominance of the Drin cascade and the projected decline in hydro output [1]. Reservoir operation has a long literature of its own [4]. Stochastic dual dynamic programming (SDDP) [6] remains the reference method for multi-stage hydrothermal scheduling under inflow uncertainty. We use a simpler two-stage structure that is adequate for investment screening (Section 4), and we note SDDP as a natural extension.

Renewable capacity expansion and open models. Linear capacity-expansion models that co-optimise investment and hourly dispatch are standard tools [7]. PyPSA [8] and its European application PyPSA-Eur [9] established open, reproducible practice for such models. Studies with them have shown, for example, the value of transmission cooperation in highly renewable systems [10]. Weather-derived capacity factors from reanalysis are widely used when measured output

is unavailable. They require care, however, because reanalysis wind speeds are biased and need correction at the site level [11–14].

Flexibility and storage. Reviews of flexibility options [15, 16] distinguish short-duration resources (batteries, demand response) from long-duration ones (reservoir hydro, interconnection). The value of short-duration storage depends strongly on the temporal resolution of the model: coarse time steps average away the intra-day variability that storage monetises [17]. Representative-period methods reduce computation but must treat seasonal storage carefully [18]. Projected storage costs continue to fall [19]. The market value of variable renewables declines as their penetration rises [20], which is directly relevant to solar-heavy expansion in a small system.

Stochastic and risk-sensitive planning. Two-stage stochastic programming [21] separates here-and-now investment from wait-and-see operation. The value of the stochastic solution (VSS) and the expected value of perfect information (EVPI) quantify, respectively, the benefit of modelling uncertainty and the cost of not knowing the future [21, 22]. Conditional value at risk (CVaR) admits a linear representation [23, 24] and is widely used in electricity decision problems [25]. Reliability is commonly priced through a value of lost load (VOLL), whose empirical estimates vary by an order of magnitude across countries and customer groups [26]. This motivates the sensitivity analysis reported below.

Gap. We are not aware of an open, reproducible model of Albania that combines (i) a physically explicit reservoir water balance, (ii) independent battery power and energy investment, (iii) an interconnector that is limited rather than infinite, (iv) two-stage stochastic investment with CVaR, and (v) a marginal-value analysis of flexibility with dual–finite-difference reconciliation. This work offers such a model as a transparent research prototype.

3 Data and study context

Scope. The study area is Albania, represented as a single node with an aggregate interconnector to its neighbours (Fig. 1). All series are indexed in UTC. Local time (Europe/Tirane) is used only to shape behavioural demand and price patterns. Monetary values are in 2024 euros, converted from USD at 0.92 EUR/USD. Energies are in MWh, powers in MW and water volumes in hm^3.

Meteorological inputs (observed-derived). Hourly all-sky irradiance, 2-m temperature and 50-m wind speed for 2019–2023 were retrieved from the NASA POWER point API [28] for five points: two solar (Fier/Myzeqe, Vlorë), two coastal wind (Vlorë–Karaburun, Lezhë–Shëngjin) and a Tirana demand proxy. Daily bias-corrected precipitation and temperature for 2000–2023 were retrieved for four points in the Drin basin (White Drin, Black Drin, Kukës/Fierza, Koman). Open-Meteo ERA5 data for Tirana in 2021 [29] were retrieved as a cross-check only. Every dataset has a machine-readable provenance record (Table 20 in the Appendix).

PV and wind capacity factors (weather-derived). PV output follows a simplified PVWatts-type model [30]: horizontal irradiance with a constant tilt gain of 1.10, system losses of 0.86 and a temperature derating of $-0.4\%/°C$, with cell temperature estimated from a nominal operating cell temperature (NOCT) of 45°C. Wind output scales the 50-m wind speed to a 100-m hub height using a shear exponent of 0.14 and applies a generic 3/12/25 m/s power curve with 0.85 losses. The mean annual capacity factors are 0.170–0.176 for PV and 0.108–0.119 for wind across 2019–2023. The wind values are low, and are likely conservative because coarse reanalysis smooths coastal and orographic winds [11]. These are modelled availabilities, *not measured production.*

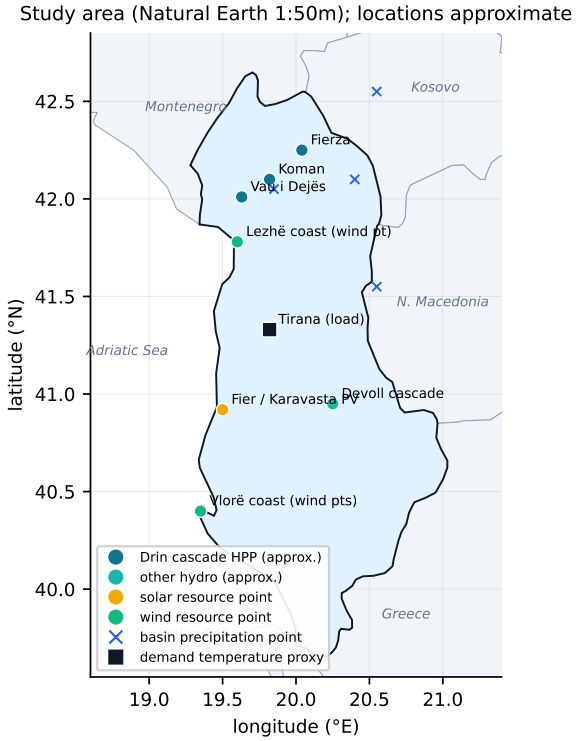


Figure 1: Study area. The Albanian boundary and its neighbours are from Natural Earth 1:50m [27]. Plant locations are approximate and shown only for orientation; the model itself has no spatial resolution. Crosses are the four NASA POWER points used to compute basin-mean precipitation for the Drin inflow reconstruction. Coloured circles are the PV and wind resource points.

Hydrology (reconstructed). Basin-mean precipitation and temperature drive a degree-day snow model [31] (threshold 0.5°C, 3 mm/°C/day). Its output feeds two parallel linear reservoirs [32] with recession constants of 4 and 45 days. The resulting runoff index is scaled so that the 2001–2023 mean inflow volume yields an *assumed* long-run Drin-cascade energy of 4.1 TWh/yr. This is consistent in magnitude with the reported mean output of about 1.33 TWh for Fierza alone, the uppermost of the three stages. Run-of-river availability for the remaining hydro fleet (about 930 MW, Table 1) is a normalised blend of fast and total runoff, capped at capacity and scaled to 2.6 TWh/yr. No gauge or plant records were accessible, so the hydrological parameters are uncalibrated assumptions. The reconstructed annual inflow index (Fig. 2d) has a 5th percentile of 0.78, a median of 0.99 and a minimum of 0.67 (2011). The reference weather year, 2022, was selected algorithmically as the 2019–2023 year closest to the median. This ranking should not be read as the observed hydrology of 2022.

Demand and prices (synthetic). ENTSO-E Transparency data require a registered token, which was not used. Hourly demand is therefore **synthetic**. It is generated from Tirana temperature with assumed heating (3.0%/°C below 16°C) and cooling (3.5%/°C above 24°C) sensitivities, an assumed diurnal and weekly shape, and AR(1) noise with a fixed seed, and it is scaled to an assumed 8.0 TWh/yr. Its peak is about 1.64 GW. Import prices are also synthetic: the mean is 105 EUR/MWh, with diurnal and seasonal shapes and multiplicative AR(1) noise. Export revenue is 85% of the import price, which represents transmission charges and market spreads. All outputs carry the label *"synthetic research demonstration data — not observed grid operations"*.

System and cost parameters. Table 1 lists all key parameters and their status. Sourced values are the Drin cascade capacities [2], the hydro fleet size [1] and the IRENA 2024 technology costs [3]. We apply an assumed regional cost multiplier of 1.25 because global weighted averages are

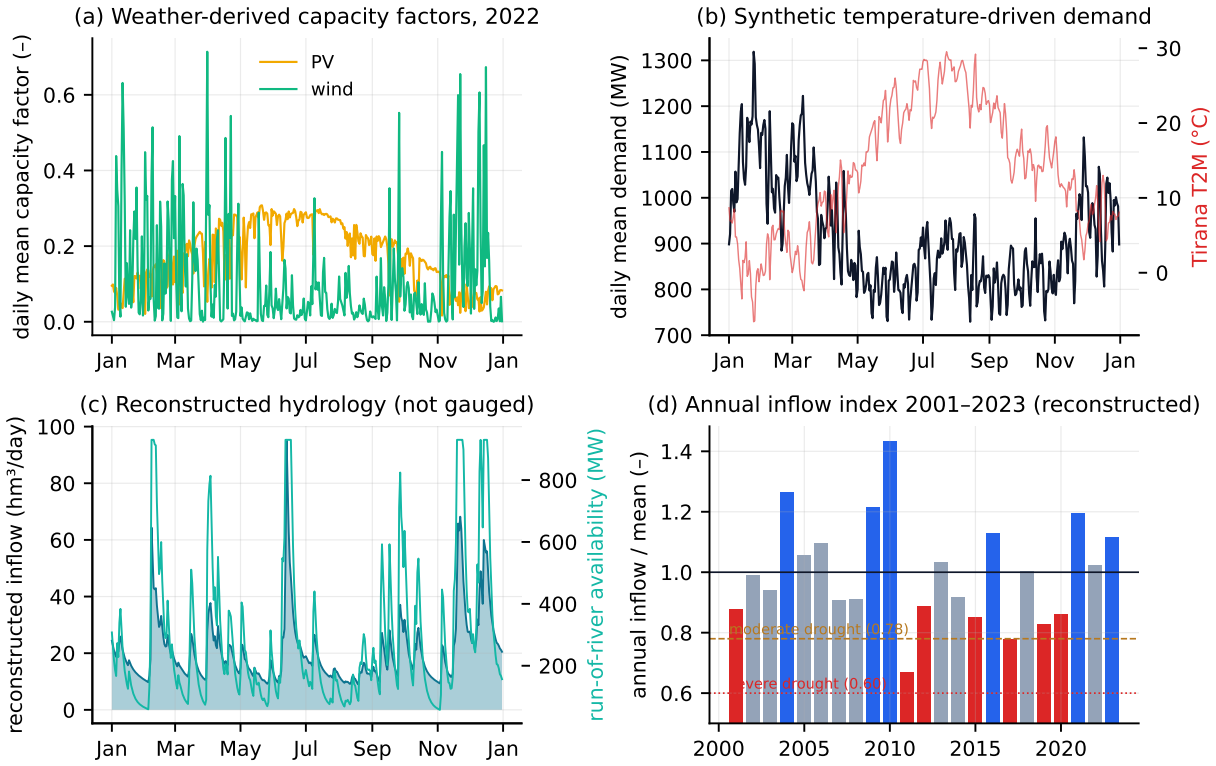


Figure 2: Model inputs for the reference year. (a) Daily mean PV and wind capacity factors derived from NASA POWER. (b) Synthetic temperature-driven demand. (c) Reconstructed Drin inflow and run-of-river availability. (d) Reconstructed annual inflow index for 2001–2023 relative to its mean, with the moderate (0.78) and severe (0.60) drought scenario levels marked.

dominated by low-cost markets. The remaining values are assumptions chosen to be of plausible magnitude, and the important ones are varied in Section 8.

Scenarios. Each scenario applies a transparent transformation to the reference-year profiles. Inflows are scaled by $(1 - d)$, demand by $(1 + g)$ and import prices by m, and interconnector availability is set to a (Table 2). The *moderate drought* ($d = 0.22$) corresponds to the 5th percentile of the reconstructed inflow index. The *severe drought* ($d = 0.40$) lies beyond the reconstructed minimum and represents a climate-stress extrapolation. The *compound shock* adds 8% demand growth, a 1.8× import-price shock and a 30% derating of transfer capacity, all assumed. For two-stage planning, five scenarios are used. Three come from binning the 23 reconstructed years (wet > 1.10, normal 0.90–1.10, dry < 0.90), with their empirical frequencies scaled to 85% of the probability mass. The other two are the severe and compound stresses, with an assumed 10% and 5%. These probabilities are partly assumed and are reported as such.

Table 1: Key system and cost parameters. "Assumption" marks values chosen for this study that were not taken from an observed Albanian dataset.

Parameter	Unit	Value	Source / status
Drin cascade turbine capacity	MW	1 350	KESH (Fierza 500, Koman 600, Vau i Dejës 250)
Aggregate head / efficiency	m / –	230 / 0.88	assumption
Energy equivalent κ	MWh/hm^3	551.5	derived
Reservoir $V^{\min}$ / $V^{\max}$	hm^3	600 / 2400	assumption
Initial volume (live fraction)	–	0.60	assumption
Minimum release	m^3/s	20	assumption
Other hydro (run-of-river aggregate)	MW	930	IRENA RRA (2,096 MW, 2018) + Moglicë 184 MW − Drin
Existing PV / wind	MW	300 / 0	assumption (order of magnitude)
Import / export NTC	MW	900 / 900	assumption
Export price ratio ρ	–	0.85	assumption
VOLL	EUR/MWh	5 000	assumption (varied 2,000–10,000)
Battery η^c, η^d / min SOC	–	0.95, 0.95 / 0.05	assumption
Battery degradation proxy	EUR/MWh	5.0	assumption
PV capex (incl. ×1.25 regional)	EUR/kW	795	IRENA RPGC 2024: USD 691/kW
Onshore wind capex (incl. ×1.25)	EUR/kW	1 197	IRENA RPGC 2024: USD 1,041/kW
Battery energy capex (incl. ×1.25)	EUR/kWh	221	IRENA RPGC 2024: USD 192/kWh
Battery power capex (incl. ×1.25)	EUR/kW	75	assumption
Discount rate / lifetimes	– / yr	0.07 / 25, 25, 15	assumption
Annualised PV / wind	kEUR/MW-yr	80.1 / 132.7	derived (CRF + FOM)
Annualised battery power / energy	kEUR/(MW, MWh)-yr	10.1 / 29.8	derived

Table 2: Climate and market scenarios. Upper block: named stress presets; lower block: five-scenario set for two-stage planning (historical-bin probabilities from the reconstructed 2001–2023 inflow index, plus an assumed 15% climate-stress allowance).

Scenario	Inflow factor	Demand factor	Import price	NTC avail.	Prob.
Normal	1.00	1.00	1.00	1.00	–
Moderate drought	0.78	1.00	1.00	1.00	–
Severe drought	0.60	1.00	1.00	1.00	–
Compound shock	0.60	1.08	1.80	0.70	–
Wet year (stoch.)	1.226	1.00	1.00	1.00	0.222
Normal (stoch.)	0.989	1.00	1.00	1.00	0.370
Dry year (stoch.)	0.823	1.00	1.00	1.00	0.259
Severe drought (stoch.)	0.600	1.00	1.00	1.00	0.100
Compound shock (stoch.)	0.600	1.08	1.80	0.70	0.050

4 Mathematical formulation

4.1 Sets, decisions and parameters

Let $\mathcal{T} = \{0, \dots, T-1\}$ be time steps of length Δt (h) and $\mathcal{S}$ the scenarios, each with probability π_s. First-stage decisions $x = (x_{pv}, x_w, x_{bp}, x_{be})$ are the installed PV and wind capacity (MW), battery power (MW) and battery energy (MWh). Each satisfies $x_k^0 \le x_k \le x_k^0 + \bar{x}_k$, where x_k^0 is the existing fleet and $\bar{x}_k$ the maximum addition. Import capacity x_{imp} (MW) and maximum reservoir volume x_{vmax} (hm^3) are modelled as fixed first-stage variables ($x_k = x_k^0$). This lets their reduced costs be read as marginal values (Section 4.5). Second-stage decisions for each (s,t) are the PV, wind and run-of-river generation g^{pv}, g^w, g^{ror}, the reservoir hydro output h, spill σ (hm^3/h), reservoir volume V (hm^3), battery charge c, discharge d and state of charge e, imports m, exports y and unserved energy u. All are non-negative.

4.2 Objective

$$\min\ w \sum_{k \in \{pv,w,bp,be\}} A_k\,(x_k - x_k^0)\ +\ \sum_s \pi_s L_s\ +\ \lambda\Big[\zeta + \frac{1}{1-\alpha}\sum_s \pi_s \phi_s\Big], \tag{1}$$

$$L_s = \frac{1}{1000}\sum_t \Delta t\,\Big(p^m_{s,t} m_{s,t} - \rho\, p^m_{s,t} y_{s,t} + \mathrm{VOLL}\,u_{s,t} + c^h h_{s,t} + c^{\deg} d_{s,t}\Big) \quad \text{(kEUR)}, \tag{2}$$

$$\phi_s \ge L_s - \zeta, \qquad \phi_s \ge 0, \qquad \zeta \in \mathbb{R}. \tag{3}$$

Here $A_k = C_k[\,r(1+r)^{n_k}/((1+r)^{n_k}-1) + f_k\,]$ is the annualised cost per unit, combining the capital recovery factor with fixed O&M as a share f_k of overnight cost C_k. The factor $w = T\Delta t/8760$ scales annual costs to the modelled horizon. Existing capacity is sunk and therefore not charged. The remaining parameters are the import price p^m, the export-to-import price ratio ρ, hydro variable O&M c^h and a battery throughput degradation proxy $c^{\deg}$. At the optimum, ζ equals the value at risk VaR_α of the operating-cost distribution, and the bracketed term equals CVaR_α [23, 24]. With $\lambda = 0$ the model is risk neutral; with $\lambda > 0$ it penalises the expected cost of the worst $(1-\alpha)$ tail. Investment cost is identical across scenarios, so the risk term acts on operating cost.

4.3 Constraints

Energy balance. For every (s,t), whose dual is the system marginal cost:

$$g^{pv}_{s,t} + g^w_{s,t} + g^{ror}_{s,t} + h_{s,t} + d_{s,t} + m_{s,t} + u_{s,t} = D_{s,t} + c_{s,t} + y_{s,t}. \tag{4}$$

Curtailment is implicit: $g^{pv}_{s,t} \le f^{pv}_t x_{pv}$ and $g^w_{s,t} \le f^w_t x_w$, so curtailed energy is $f^{pv}_t x_{pv} - g^{pv}_{s,t}$ and does not appear in (4). This avoids double counting. Run-of-river generation satisfies $g^{ror}_{s,t} \le R_{s,t}$, where $R_{s,t}$ is its availability.

Reservoir. With the energy equivalent $\kappa = \rho_w g H \eta \cdot 10^6/3.6 \times 10^9$ (MWh/hm^3),

$$V_{s,t+1} = V_{s,t} + \Delta t\,(I_{s,t} - h_{s,t}/\kappa - \sigma_{s,t}), \qquad V^{\min} \le V_{s,t} \le x_{vmax}, \tag{5}$$

$$0 \le h_{s,t} \le P^h, \qquad h_{s,t}/\kappa + \sigma_{s,t} \ge r^{\min}, \tag{6}$$

$$V_{s,0} = V^0_s, \qquad V_{s,T} \ge V^T_s. \tag{7}$$

With $H = 230$ m and $\eta = 0.88$, $\kappa = 551.6$ MWh/hm^3. The fixed-head assumption makes the model linear; a variable-head model would be bilinear in V and q and is left for future work. In annual runs $V^T_s = V^0_s$, which closes the seasonal cycle. For shorter windows, V^0_s and V^T_s come from a coarse (6-hourly) full-year run of the same scenario. We call this the *nested boundary*. It prevents an isolated week from either exhausting or hoarding seasonal storage.

Battery.

$$e_{s,t+1} = e_{s,t} + \Delta t(\eta^c c_{s,t} - d_{s,t}/\eta^d), \qquad \varepsilon\, x_{be} \le e_{s,t} \le x_{be}, \tag{8}$$

$$0 \le c_{s,t},\, d_{s,t} \le x_{bp}, \qquad e_{s,0} = s_0 x_{be},\;\; e_{s,T} \ge e_{s,0}, \qquad x_{be} \le \delta^{\max} x_{bp}. \tag{9}$$

Simultaneous charging and discharging is not excluded by binary variables. Because $\eta^c\eta^d < 1$ and $c^{\deg} > 0$, the LP has no incentive to do it unless it must dissipate energy, and the quantity is reported ex post for every run.

Interconnection, shortage and budget.

$$0 \le m_{s,t} \le a_s\, x_{imp}, \qquad 0 \le y_{s,t} \le Y, \qquad 0 \le u_{s,t} \le D_{s,t}, \qquad \sum_k C_k(x_k - x_k^0) \le B. \tag{10}$$

The import limit is an aggregate net transfer capacity scaled by scenario availability a_s. It is not a power-flow model.

4.4 Representative weeks for two-stage planning

Hourly resolution matters for batteries, and seasonal coupling matters for the reservoir [17, 18]. To retain both, investment and stochastic problems use eight hourly weeks per scenario, starting on days 7, 52, ..., 325 (a systematic sample about every 45.6 days). Each (scenario, week) pair is a sub-problem with probability $\pi_s/8$. It starts at the volume $V^0_{s,w}$ of a coarse (6-hourly) full-year run of scenario s with the existing fleet. Its end volume is not fixed. Instead it is valued at the marginal water value $\omega_{s,w}$ of that run, the negative dual of the water-balance constraint (5) at the end of the week (kEUR/hm^3):

$$L_{s,w} \leftarrow L_{s,w} + \omega_{s,w}\,(\bar{V}^T_{s,w} - V_{s,w,T}), \tag{11}$$

where $\bar{V}^T_{s,w}$ is the coarse run's end-of-week volume. This is a first-order (one-cut) future-cost function, in the spirit of the water values used in stochastic dual dynamic programming [6]. Water used beyond the coarse plan is charged at its opportunity cost, and water saved is credited. The battery is cyclic within each week. CVaR excess variables are defined per scenario, not per week: $\phi_s \ge \sum_w \frac{1}{8} L_{s,w} - \zeta$. The boundaries do not depend on the plan, so the stochastic programme, the deterministic plan's evaluation, the wait-and-see problems and all frontier evaluations share identical seasonal information. This guarantees a consistent comparison. Section 7.9 compares this approximation with full-year chronological evaluation. It also compares it with an earlier hard-boundary variant ($V_{s,w,T} \ge \bar{V}^T_{s,w}$), which we found created spurious scarcity.

4.5 Marginal value of flexibility

For a capacity x_k fixed by bounds, HiGHS returns the reduced cost $\bar{d}_k = c_k - (A^\top y)_k$. The marginal operating value is $\mathrm{MV}_k = -\partial\mathbb{E}[L]/\partial x_k = c_k - \bar{d}_k - (A^\top_{\mathrm{1st}} y)_k$. The last term removes contributions from first-stage rows such as the budget and duration constraints. The optimal operating cost is a convex, piecewise-linear function of x_k [33]. Hence MV_k is a subgradient: it lies between the finite-difference value of adding the next increment, Δ^+_k, and that of removing the last one, Δ^-_k. Duals are therefore local and may be non-unique under degeneracy. At zero capacity (for example $x_{bp} = 0$, or $x_{be} > 0$ with $x_{bp} = 0$) the corner is degenerate and the dual is uninformative. We therefore report finite differences alongside duals and flag corner cases.

5 Computational implementation

Software. The model is implemented in Python 3.13 (NumPy, pandas, SciPy). The primary engine is a vectorised sparse LP builder that assembles the constraint matrix in coordinate form and

passes it to HiGHS [34] (highspy 1.15) as a column-wise model. An independent implementation of the deterministic model in PyPSA 1.4 [8], with linopy and HiGHS, serves as a cross-check. In that version the reservoir is a `StorageUnit` whose state of charge is $(V - V^{\min})\kappa$ and whose inflow is $I\kappa$. The battery is a `Store` with charger and discharger `Links`, exports are a generator with negative output, and shortage is a VOLL-priced generator. A FastAPI service exposes the same functions to the interactive application (Fig. 3).

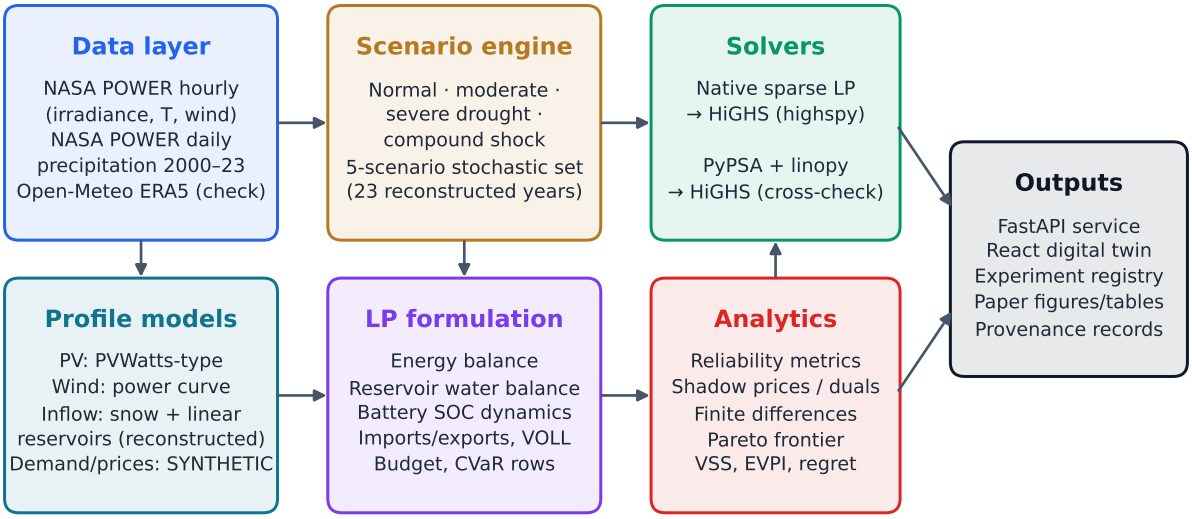


Figure 3: HYDROFLEX AI architecture, from public data through the profile models, the scenario engine, the LP formulation, the solvers and the analytics to the outputs.

Solver configuration. Single-scenario dispatch problems use HiGHS' default strategy. Chronological multi-scenario and annual investment problems use the interior-point method with crossover. Representative-week problems use dual simplex: the first-stage capacity columns appear in every hourly row, producing dense columns that caused interior-point iterations to stall (Table 18). Live API runs are limited to 8,760 steps for dispatch, 2,920 steps for investment and 7,300 scenario-steps for stochastic runs, with solver time limits and a per-client rate limit.

Verification. The automated test suite (67 Python tests; 42 frontend unit tests and 3 browser tests) checks: energy balance; battery energy conservation; reservoir water conservation; capacity, interconnector and budget limits; non-negativity of unserved energy; zero-solar, zero-battery, zero-import and zero-demand behaviour; visible shortages under insufficient supply; the non-increase of unserved energy as VOLL rises; correct reporting of an infeasible configuration; reconciliation of the objective with its components (including grouped CVaR and the terminal water value); non-negative VSS; and the subgradient relation between duals and finite differences. It also tests the API contract and data provenance completeness. The native model is compared with PyPSA on five cases (Table 19).

Reproducibility. `python -m hydroflex.experiments.run_all` executes every experiment. It writes machine-readable results to `results/experiments/`, records software versions, seeds and timings in a manifest, and regenerates all tables and figures in this paper. Synthetic series use fixed seeds (`20260101+year`).

6 Experimental design

Table 3 summarises the ten experiments. Experiments A–G and J are run as full-year (8,760 h) hourly dispatch with fixed fleets. Experiments H and I are investment problems solved with the representative-week formulation. Every strategy in the investment comparison is *evaluated* with the same representative-week model on the five-scenario set, so that strategies are compared on equal terms. A resolution study (1, 2, 3 and 6 h) quantifies the bias that coarse time steps introduce into storage value.

Table 3: Experiments and hypotheses. All hypotheses are tested numerically in the pipeline; whether each is supported is reported in Section 7.

ID	Experiment	Hypothesis
A	Reference operation	With reference inflows the existing fleet serves demand without shortage, using the interconnector in both directions.
B	Drought (moderate, severe)	Drought is absorbed mainly through higher net imports and cost; shortages do not occur while the interconnector is fully available.
C	Solar without batteries	Import reduction per added MW of PV declines as PV grows (diminishing returns).
D	Batteries without new renewables	Battery operating value is modest relative to annualised cost where reservoir hydro already provides flexibility.
E	Coordinated portfolio	PV and storage are complements: combined operating savings exceed the sum of the separate additions.
F	Import price shock / derating	Price shocks raise cost but not shortage; shortages require a loss of import availability.
G	Demand growth	With halved import availability, unserved energy rises non-linearly with demand growth.
H	Budget-constrained investment	Under a binding budget the optimiser favours the technology with the highest operating value per euro.
I	Deterministic vs stochastic	A stochastic, risk-aware plan reduces expected cost and tail risk relative to planning for a normal year.
J	Marginal value of flexibility	Dual values lie within the finite-difference interval wherever the LP is non-degenerate.

7 Results

Every number in this section is read from the experiment files in `results/experiments/`. All solves terminated with HiGHS status *Optimal*. The maximum energy-balance residual was below 10^{-4} MW and the water-balance residual below 10^{-6} hm^3 in every run, and no simultaneous battery charging and discharging was detected. Table 4 records whether each hypothesis was supported.

Table 4: Outcome of the numerical hypothesis tests (generated from `ref-summary.json`).

ID	Experiment	Hypothesis
A	Reference operation, normal conditions	supported
B	Hydropower reduction under drought	supported
C	Solar without batteries	supported
D	Batteries without new renewables	supported
E	Coordinated solar, wind, hydro and storage	supported
F	Import price shock and interconnector derating	supported
G	Demand growth stress test	supported
H	Investment planning under capital constraints	descriptive
I	Deterministic versus stochastic planning	not supported
J	Marginal value of flexibility	supported
I2	When stochastic planning changes the decision	supported
K	Representative weeks vs full-year chronology	supported
R	Temporal-resolution sensitivity	supported

7.1 Reference operation and drought stress (Experiments A, B, F)

Table 5 summarises the full-year hourly runs with the existing fleet. Under normal conditions the synthetic demand of 8.00 TWh is fully served. Hydro supplies 6.77 TWh, and the interconnector is used in both directions: 2.00 TWh of imports and 1.23 TWh of exports, for net imports of 0.77 TWh. The reservoir works as a seasonal and daily arbitrage store. It imports in low-price hours and exports in high-price hours, so annual operating cost is only 40.8 MEUR. Drought shifts this balance. Hydro output falls to 5.28 TWh with moderate drought and 4.06 TWh with severe drought. Net imports rise to 2.26 and 3.48 TWh, and operating cost rises to 192.7 and 324.5 MEUR/yr. The number of hours with imports at the interconnector limit grows from 91 to 203 and 380. **No energy is unserved in either drought case**. While the 900 MW interconnector is fully available, drought in this model is a *financial and import-dependence* risk rather than a reliability risk. Figures 4 and 5 show how the dispatch and reservoir trajectories change.

The compound shock combines severe drought with 8% demand growth, 1.8× import prices and 70% transfer availability. Operating cost rises to 727.1 MEUR/yr, imports sit at the derated limit for 4,220 h, and the mean shadow price is 209 EUR/MWh. Even so, demand is still fully served over the year.

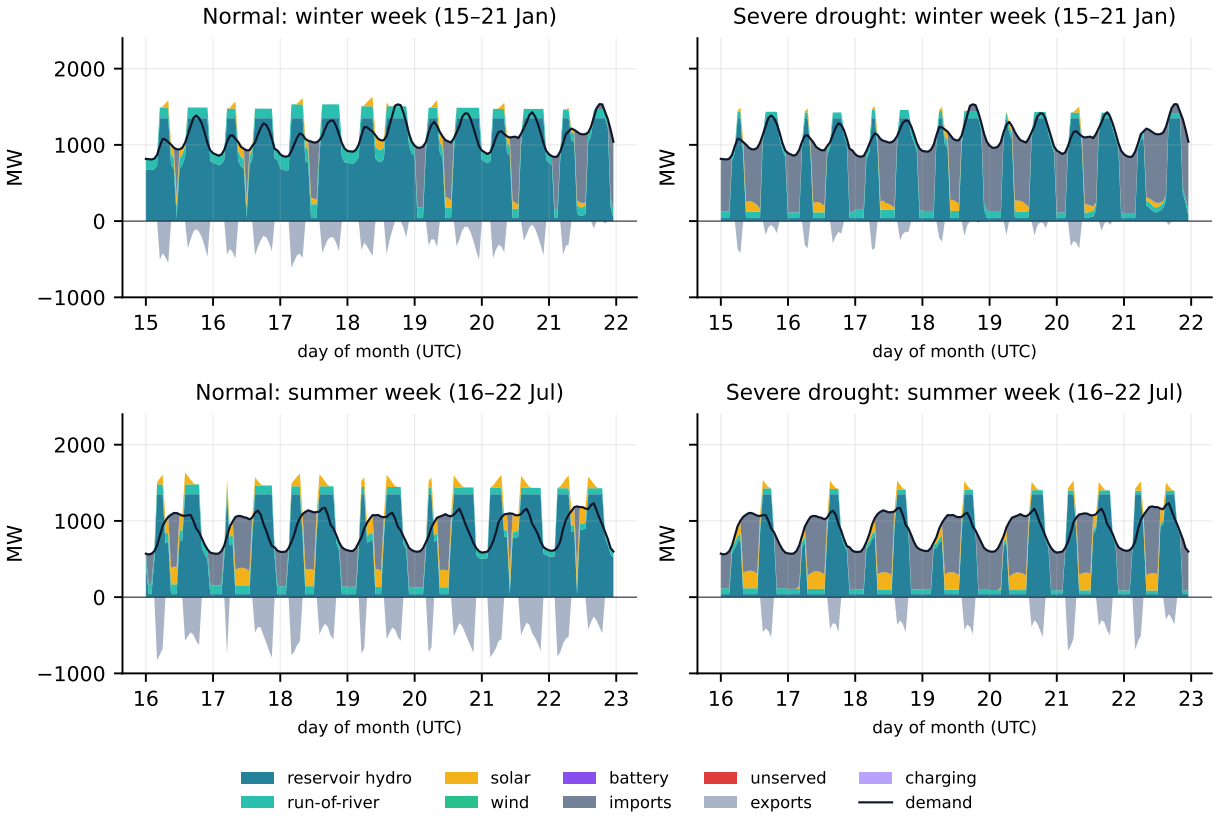


Figure 4: Hourly dispatch in a winter and a summer week under normal conditions and severe drought (existing fleet, full-year optimisation). Exports and battery charging are drawn below zero. In drought the reservoir covers fewer hours and imports fill the evening peaks.

7.2 Where reliability fails: interconnector availability and demand growth (Experiments F, G)

A uniform import-price shock leaves dispatch unchanged, because relative prices stay the same. Annual cost then rises linearly, from 324.5 MEUR at 1× to 968.4 MEUR at 3× (severe drought), with zero unserved energy (Table 6). Shortage appears only when import *availability* falls. Under

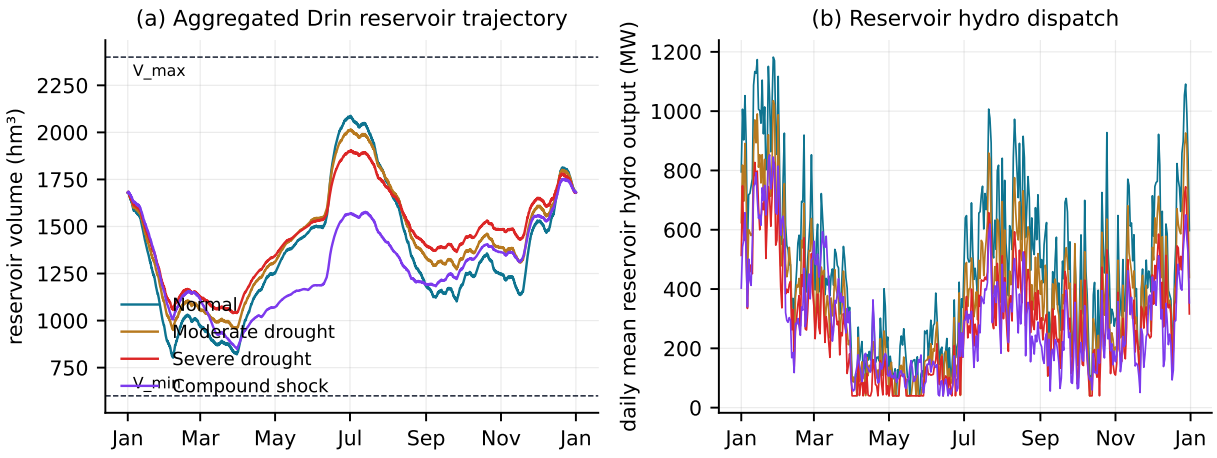


Figure 5: (a) Volume trajectory of the aggregated Drin reservoir and (b) daily reservoir hydro output under the four presets (full-year hourly optimisation with $V_T = V_0$). The minimum operating volume is not reached in any preset. Drought lowers releases rather than draining the reservoir, because the cyclic terminal condition forces the deficit to be covered by imports.

Table 5: Annual hourly dispatch (8760 h, reference weather year). Energies in TWh, ENS in MWh, operating cost (imports − exports + penalties + variable O&M) in MEUR/yr, mean shadow price in EUR/MWh. Existing fleet unless stated.

Case	Demand	Hydro	PV+wind	Imports	Exports	ENS	Opex	Price
Normal	8.00	6.77	0.46	2.00	1.23	0	40.8	97.2
Moderate drought	8.00	5.28	0.46	2.91	0.66	0	192.7	99.9
Severe drought	8.00	4.06	0.46	3.79	0.32	0	324.5	102.2
Compound shock	8.64	4.06	0.46	4.20	0.09	0	727.1	209.3
Coordinated fleet, normal	8.00	6.77	2.28	1.22	2.20	0	-133.1	93.5
Coordinated fleet, severe	8.00	4.06	2.28	2.63	0.90	0	139.4	99.4

severe drought with 8% demand growth, availability of 75% still serves all demand. At 50%, 311 GWh are unserved over 999 h with a peak shortfall of 811 MW. At 35% and 25% availability, unserved energy rises to 1.40 and 2.16 TWh. These shortages are *energy*-limited, not power-limited. With 50% availability the annual supply that can be delivered (hydro + PV + 450 MW of imports × 8,760 h) falls short of demand. Consistent with this, a 400 MW / 1,600 MWh battery reduces unserved energy at 50% availability by only 13% (311 to 270 GWh). At VOLL = 5,000 EUR/MWh even this partial reduction lowers annual cost by 199.8 MEUR, well above the battery's 51.7 MEUR/yr annualised cost. At 35% and 25% availability the reduction is 1–2% and the saving 97 and 59 MEUR/yr. Demand growth shows the same threshold behaviour (Table 16, Fig. 6c). With full interconnector availability, demand growth of up to 30% is absorbed by imports. With 50% availability, unserved energy is 0, 456, 1,201 and 1,969 GWh at growth of 0, 10, 20 and 30%, so loss-of-load hours rise from 0 to 4,088. In this model, therefore, the reliability of a hydro-dependent system in drought is governed mainly by the availability of interconnection. This matters because regional droughts can depress neighbours' export capacity at the same time.

Table 6: Experiment F: import price shocks and interconnector derating under severe drought (annual, hourly).

Case (severe drought)	Imports (TWh)	Net imp. (TWh)	LOLH (h)	ENS (MWh)	Cost (MEUR/yr)
Import price ×1.0	3.79	3.48	0	0	324.5
Import price ×1.5	3.79	3.48	0	0	485.5
Import price ×2.0	3.79	3.48	0	0	646.5
Import price ×2.5	3.79	3.48	0	0	807.4
Import price ×3.0	3.79	3.48	0	0	968.4
NTC avail. 100%, +8% demand	4.33	4.12	0	0	391.7
NTC avail. 75%, +8% demand	4.24	4.12	0	0	400.7
NTC avail. 50%, +8% demand	3.81	3.81	999	310 714	1 957.2
NTC avail. 35%, +8% demand	2.71	2.71	3 391	1 401 990	7 298.1
NTC avail. 25%, +8% demand	1.95	1.95	4 310	2 162 994	11 022.9
NTC 50%, +8% dem., battery 400/1600	3.86	3.86	816	270 120	1 757.4
NTC 35%, +8% dem., battery 400/1600	2.74	2.74	3 293	1 382 198	7 200.9
NTC 25%, +8% dem., battery 400/1600	1.97	1.97	4 267	2 151 012	10 964.1

7.3 Solar, storage and coordination (Experiments C, D, E)

Solar without storage. Adding PV reduces gross imports with diminishing returns (Table 7). Under normal conditions the reduction per added MW is 1,074, 898 and 699 MWh/MW for +500, +1,000 and +1,500 MW. Under severe drought it is larger (1,406, 1,253 and 1,022 MWh/MW), because a drier system exports less and imports more at the margin. Operating savings fall only slightly per MW (138 to 131 kEUR/MW-yr in the normal year), because surplus midday PV can still be exported at 85% of the import price. Curtailment therefore stays small: at most 21 GWh, at +1,500 MW in the normal year. This result rests on the assumed export capacity and price ratio (Section 8). The annualised cost of PV is 80 kEUR/MW-yr, so under these assumptions every tested PV addition reduces total cost.

Batteries without new renewables. A battery earns only intra-day arbitrage. The annual operating savings are 5.5, 9.8 and 13.8 MEUR (normal) and 7.0, 12.7 and 17.9 MEUR (severe drought) for 200/800, 400/1,600 and 800/3,200 MW/MWh. Their annualised costs are 25.8, 51.7 and 103.3 MEUR/yr, so arbitrage alone does not recover them. The value per unit is higher in drought, because the import price, not the cheap reservoir, sets the marginal cost in more hours. Batteries also *increase* gross imports: by 0.23–0.60 TWh in the normal year and similar amounts in the severe-drought year. They do this by storing low-price imports for high-price hours. "Reducing imports" and "reducing cost" are therefore different objectives for storage.

Coordination. Combining PV (+1,000 MW) with a 400 MW / 1,600 MWh battery saves 146.9 MEUR/yr in the normal year. That is 1.75 MEUR more than the sum of the separate

savings (135.4 + 9.8). In severe drought the corresponding synergy is 1.82 MEUR (Table 8). The synergy is positive but small, because an export route and the reservoir already absorb most PV surplus. The battery removes the remaining 2.2 GWh of PV curtailment.

Table 7: Experiment C: PV additions without storage (annual, hourly).

Case	Imports (TWh)	Net imp. (TWh)	Curtail. (GWh)	Batt. cycles	ENS (MWh)	Opex (MEUR/yr)
Normal, PV 300 MW	2.00	0.77	0.0	0.0	0	40.8
Normal, PV 800 MW	1.46	0.00	0.0	0.0	0	-28.3
Normal, PV 1300 MW	1.10	-0.76	2.2	0.0	0	-94.6
Normal, PV 1800 MW	0.95	-1.51	20.7	0.0	0	-156.3
Severe, PV 300 MW	3.79	3.48	0.0	0.0	0	324.5
Severe, PV 800 MW	3.09	2.71	0.0	0.0	0	252.7
Severe, PV 1300 MW	2.54	1.94	0.0	0.0	0	182.8
Severe, PV 1800 MW	2.26	1.18	2.3	0.0	0	117.0

Table 8: Experiment E: coordinated portfolios (fixed fleets, annual hourly dispatch). Battery = 400 MW / 1600 MWh.

Case	Imports (TWh)	Net imp. (TWh)	Curtail. (GWh)	Batt. cycles	ENS (MWh)	Opex (MEUR/yr)
Normal, baseline	2.00	0.77	0.0	0.0	0	40.8
Normal, pv+1000	1.10	-0.76	2.2	0.0	0	-94.6
Normal, batt400/1600	2.42	0.84	0.0	401.4	0	31.0
Normal, pv+1000&batt	1.35	-0.70	0.0	390.0	0	-106.2
Normal, pv+1000&wind300	0.99	-1.05	2.8	0.0	0	-121.7
Normal, pv+1000&wind300&batt	1.22	-0.99	0.0	385.1	0	-133.1
Severe, baseline	3.79	3.48	0.0	0.0	0	324.5
Severe, pv+1000	2.54	1.94	0.0	0.0	0	182.8
Severe, batt400/1600	4.20	3.56	0.0	463.8	0	311.8
Severe, pv+1000&batt	2.84	2.02	0.0	470.3	0	168.3
Severe, pv+1000&wind300	2.34	1.66	0.0	0.0	0	153.8
Severe, pv+1000&wind300&batt	2.63	1.74	0.0	468.5	0	139.4

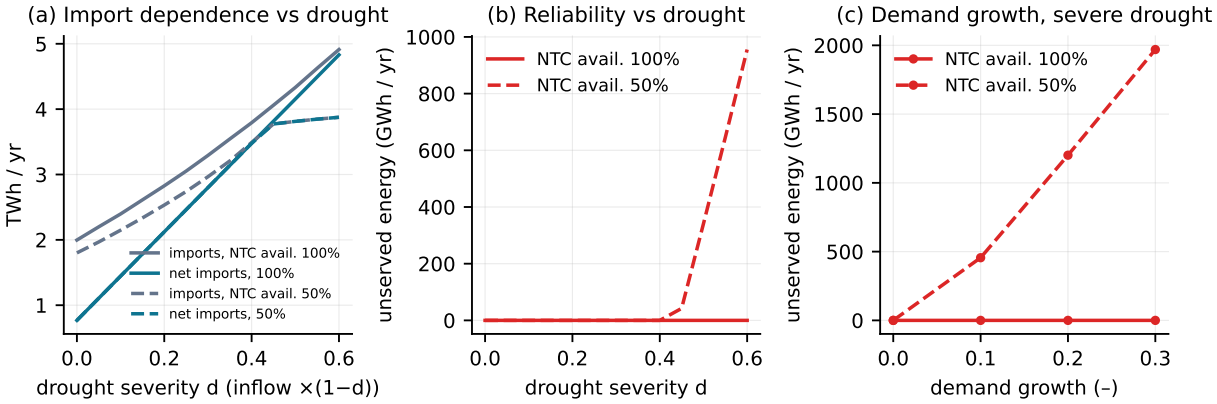


Figure 6: (a) Gross and net imports and (b) unserved energy as functions of drought severity d (inflows scaled by $1-d$, existing fleet) with full (solid) and halved (dashed) interconnector availability. (c) Unserved energy under severe drought as demand grows. Shortage appears only when transfer availability is restricted.

7.4 Marginal value of flexibility (Experiment J)

Table 9 and Fig. 8 report finite-difference and dual-based marginal values around a base fleet with PV 300 MW, a 200 MW / 800 MWh battery and 900 MW of import capacity. In every non-degenerate case the dual lies within the finite-difference interval $[\Delta^+, \Delta^-]$, as LP theory predicts. Four results stand out:

- **PV** has an operating value of 140 kEUR/MW-yr (normal), 146 (severe drought) and 314 (compound shock), against an annualised cost of 80 kEUR/MW-yr.

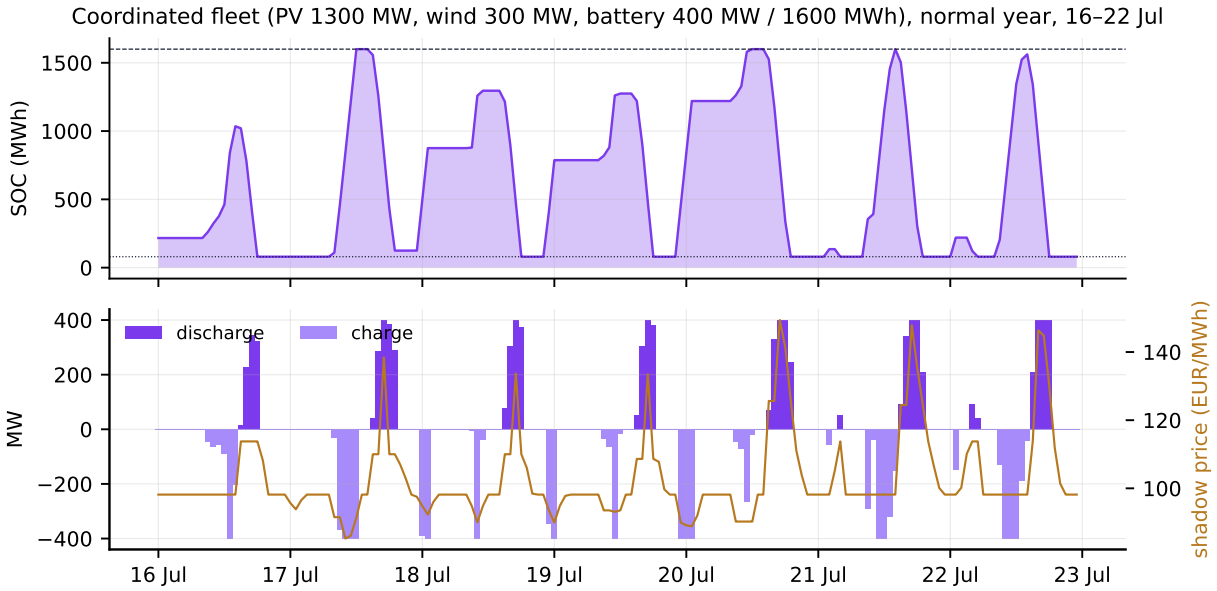


Figure 7: Battery operation in a summer week (coordinated fleet: PV 1,300 MW, wind 300 MW, battery 400 MW / 1,600 MWh). The battery charges around midday when PV depresses the shadow price and discharges in the evening peak: roughly one cycle per day (385 equivalent full cycles per year).

- **Wind** (CF $\approx$ 0.11 from coarse reanalysis) is worth 93–99 kEUR/MW-yr against 133 kEUR/MW-yr in normal and drought years, and 202 under the compound shock. The wind dual at zero capacity is flagged as a degenerate corner, and the right finite difference is reported instead.
- **Battery power and energy are not substitutes.** With 800 MWh installed, an extra MW of discharge power is worth 12.7–15.5 kEUR/yr in normal and drought years, above its 10.1 kEUR/yr annualised cost. An extra MWh is worth only 2.3–4.1 kEUR/yr, against 29.8. Shorter-duration batteries are therefore closer to break-even than the 4-hour configuration.
- **Import capacity** is worth 2 kEUR/MW-yr in a normal year, 9 in severe drought and 175 under the compound shock. This is the largest increase of any resource, and it confirms that interconnection is the critical flexibility resource under compound stress. Additional reservoir *volume* has zero marginal value in all cases: the volume bound never binds, and the energy limit comes from inflow rather than storage size.

7.5 Budget-constrained investment (Experiment H)

For the normal year, the deterministic optimum spends every budget level from 100 to 1,500 MEUR entirely on PV: 126, 315, 629, 944, 1,258 and 1,888 MW. At 2,000 MEUR the PV bound (+2,000 MW) binds and the budget does not. Neither wind nor storage enters at any budget. The budget shadow price falls from 0.070 to 0.024 EUR of annual saving per euro of capital as PV grows. These are diminishing but still positive returns, and they are consistent with the PV marginal value of Experiment J (140 kEUR/MW-yr against 80 kEUR/MW-yr of annualised cost). The result reflects three assumptions: exports absorb midday surplus at 85% of the import price, there is no reserve requirement, and wind capacity factors are about 0.11 from coarse reanalysis. Section 8 tests the first of these.

7.6 Deterministic versus stochastic planning (Experiment I)

The two-stage programme has five scenarios, eight hourly weeks each, a 500 MEUR budget and CVaR at $\alpha = 0.9$. For every risk weight tested ($\lambda \in \{0, 0.25, 1, 4\}$) it selects the *same* portfolio

Table 9: Marginal operating value of flexibility (kEUR per unit per year) around the base fleet (PV 300 MW, battery 200 MW/800 MWh, NTC 900 MW, $V^{\max}$ 2400 hm^3). Δ^+: value of adding the next increment; Δ^-: value of the last increment. Ann. cost: annualised cost of one more unit.

Resource	Scenario	Unit	Δ^+	Δ^-	Dual	Dual valid	Ann. cost
battery energy	normal	MWh	2.3	3.0	2.6	yes	29.8
battery power	normal	MW	12.7	16.2	14.5	yes	10.1
solar pv	normal	MW	139.7	140.0	139.8	yes	80.1
wind	normal	MW	92.8	–	92.9	no	132.7
import capacity	normal	MW	1.7	2.7	2.1	yes	30.0
reservoir volume	normal	hm^3	0.0	0.0	0.0	yes	–
battery energy	severe drought	MWh	3.1	4.1	3.3	yes	29.8
battery power	severe drought	MW	15.5	20.3	18.6	yes	10.1
solar pv	severe drought	MW	145.6	146.1	145.8	yes	80.1
wind	severe drought	MW	99.2	–	99.3	no	132.7
import capacity	severe drought	MW	7.5	11.2	9.3	yes	30.0
reservoir volume	severe drought	hm^3	0.0	0.0	0.0	yes	–
battery energy	compound shock	MWh	4.1	5.4	4.6	yes	29.8
battery power	compound shock	MW	8.2	11.7	10.0	yes	10.1
solar pv	compound shock	MW	314.2	322.1	318.1	yes	80.1
wind	compound shock	MW	201.8	–	202.5	no	132.7
import capacity	compound shock	MW	153.5	197.4	174.5	yes	30.0
reservoir volume	compound shock	hm^3	0.0	0.0	0.0	yes	–

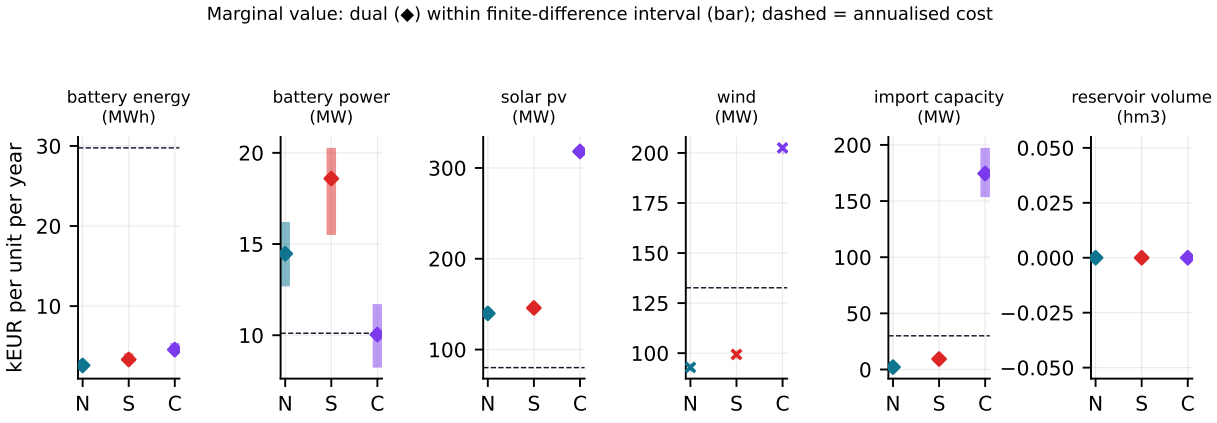


Figure 8: Marginal operating value of each flexibility resource in the normal (N), severe-drought (S) and compound-shock (C) cases. Bars show the finite-difference interval $[\Delta^+, \Delta^-]$, diamonds the dual value (crosses mark degenerate corners), and the dashed line the annualised cost of one more unit.

as the deterministic normal-year plan: +629 MW of PV, with no wind and no storage (Table 17). The same holds at a 1,500 MEUR budget. Each wait-and-see problem also chooses this plan, so the value of the stochastic solution and the expected value of perfect information are both zero to solver precision (Table 10). $CVaR_{0.9}$ of operating cost is 397 MEUR/yr for both plans. Hypothesis I is therefore **not supported** under the base assumptions. PV is a *no-regret* investment in every scenario. Its value is set mainly by the import price, and it rises under stress. The budget binds before any resource whose value depends on the scenario becomes competitive. The stochastic machinery still answers question 6 of the study: under these assumptions, deterministic planning for an average year is not penalised. One qualification matters. None of the five planning scenarios, the compound shock included (70% availability), reaches the import-availability regime of $\leq$50% in which shortages occur and batteries become valuable (Sections 7 and 8). A scenario set with even a small probability of that regime could change the plan, and this is a priority for future work.

Table 10: Two-stage stochastic versus deterministic planning (budget 500 MEUR, 8 representative weeks, water-value boundaries). E[cost]: expected total annualised cost (MEUR/yr) on all five scenarios; CVaR of operating cost (MEUR/yr). Risk-adjusted VSS = 0.00 MEUR/yr; EVPI is reported only for $\lambda = 0$.

Plan	+PV	+Wind	+B MW	+B MWh	E[cost]	$CVaR_{0.9}$
Stochastic (RP), λ=1	629	0	0	0	70.9	397.3
Deterministic (normal year)	629	0	0	0	70.9	397.3

7.7 When does stochastic planning matter? (Experiment I2)

To find the conditions under which uncertainty changes the decision, we repeated the comparison with export capacity limited to 300 MW. Surplus PV must then be stored, absorbed by the reservoir or curtailed. We also raised the budget, so that PV saturates before the budget binds (Table 11). With 900 MW of export capacity the stochastic and deterministic plans remain identical at every risk weight (VSS = 0). With restricted exports and risk-neutral planning ($\lambda = 0$), the stochastic plan adds slightly more PV than the deterministic one (+26 MW at 1,000 MEUR, +75 MW at 1,500 MEUR), and the VSS is only 0.2–0.3 MEUR/yr. With restricted exports, a 1,500 MEUR budget and $\lambda = 1$, the decision changes qualitatively. The risk-averse plan adds 1,520 MW of PV, 230 MW of *wind* and a 30 MW / 66 MWh battery. The deterministic plan adds 1,232 MW of PV only. In the representative-week model the risk-averse plan raises expected cost by 7.8 MEUR/yr and lowers $CVaR_{0.9}$ of operating cost by 79 MEUR/yr (from 292 to 213); the risk-adjusted VSS is 71.5 MEUR/yr. Full-year chronological re-evaluation of the two plans confirms the trade-off. Expected total cost is higher by 15.1 MEUR/yr (74.6 vs 59.5), $CVaR_{0.9}$ of operating cost is lower by 74.5 MEUR/yr (208.2 vs 282.7), and neither plan has any unserved energy. The hedge is mainly wind rather than storage. Once PV saturates under the export limit, wind is the cheapest remaining way to cut imports in the high-price stress scenarios. We did not decompose this mechanism further, for example by wind–PV correlation or season. Stochastic, risk-aware planning therefore matters when a no-regret resource saturates because its surplus cannot leave the system. In that regime the choice among the remaining resources depends on the tail scenarios.

7.8 Investment frontier

Figure 10 and Table 12 compare the required strategies, the budget sweep and the stochastic plans on equal terms, using the same representative-week evaluation. At 500 MEUR the strategies rank as follows by expected total annualised cost: solar (70.9 MEUR/yr; identical to the coordinated and climate-robust optima), baseline (109.4), wind (112.1) and battery-only (164.9). The battery-only plan has the highest cost and the highest imports, because arbitrage increases gross imports. Across budgets, larger PV additions lower both expected cost and imports. The non-dominated set

Table 11: Experiment I2: stochastic (RP) versus deterministic-plan evaluation (EEV) under restricted and full export capacity. Additions as PV/wind MW / battery MW/MWh; E = expected total annualised cost, $\mathrm{CVaR}_{0.9}$ of operating cost and risk-adjusted VSS in MEUR/yr (representative weeks, water-value boundaries).

Export MW	Budget	λ	Stoch. adds	Det. adds	E RP	E EEV	CVaR RP	CVaR EEV	VSS
300	1 000	0	1 258/0/0/0	1 232/0/0/0	62.0	62.1	288.1	292.0	0.2
300	1 000	1	1 258/0/0/0	1 232/0/0/0	62.0	62.1	288.1	292.0	4.1
300	1 500	0	1 307/0/0/0	1 232/0/0/0	61.8	62.1	281.1	292.0	0.3
300	1 500	1	1 520/230/30/66	1 232/0/0/0	70.0	62.1	212.7	292.0	71.5
900	1 500	0	1 888/0/0/0	1 888/0/0/0	12.9	12.9	170.2	170.2	0.0
900	1 500	1	1 888/0/0/0	1 888/0/0/0	12.9	12.9	170.2	170.2	0.0

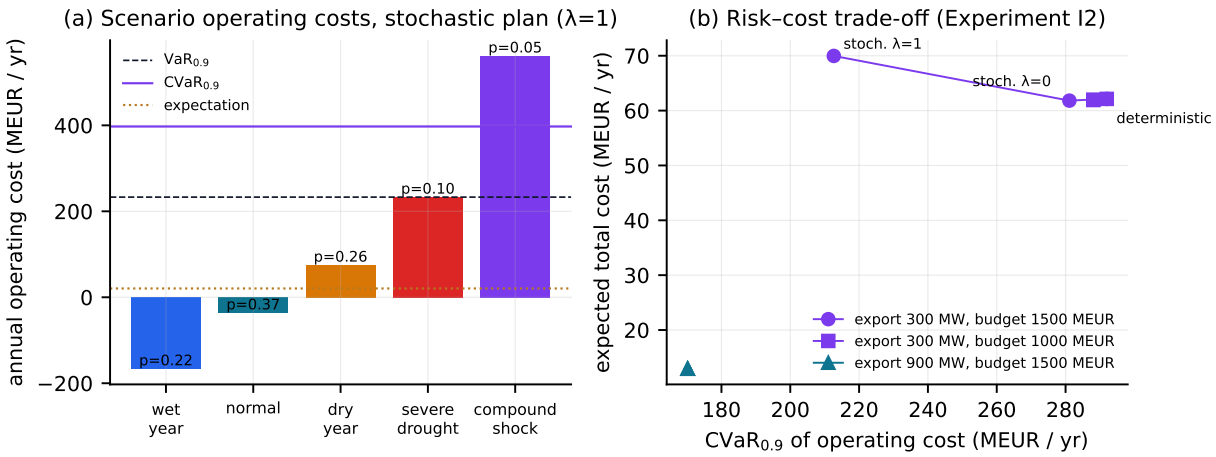


Figure 9: (a) Annualised operating cost of the reference stochastic plan in each scenario, with probabilities, $\mathrm{VaR}_{0.9}$, $\mathrm{CVaR}_{0.9}$ and the expectation. In the base case this plan is identical to the deterministic plan. (b) Expected total cost versus $\mathrm{CVaR}_{0.9}$ of operating cost for the deterministic plan and the stochastic plans ($\lambda = 0, 1$) in Experiment I2. Only with restricted exports and a large budget does risk aversion buy a material reduction in tail cost.

in (expected cost, expected imports) therefore collapses to the largest-PV plan, and the trade-off appears only in capital spent versus imports (`pareto_capex_imports` in the artefact). We report this degenerate frontier as it is, rather than constructing an artificial trade-off.

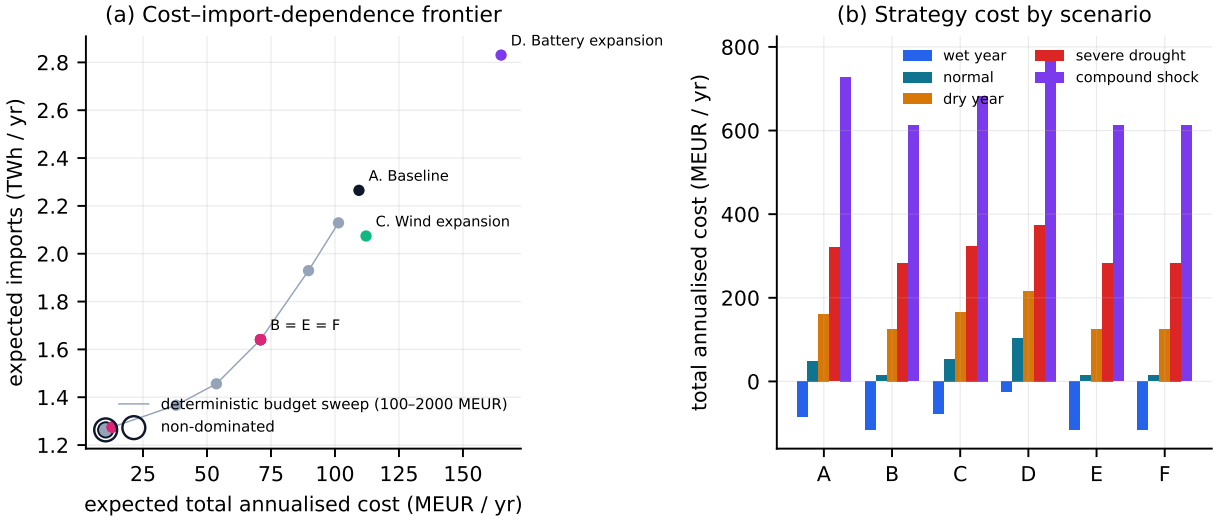


Figure 10: (a) Strategies in the plane of expected total annualised cost and expected imports, evaluated with the same representative-week model over five scenarios. Outlined points are non-dominated. (b) Total annualised cost of the required strategies A–F in each scenario.

Table 12: Investment strategies evaluated on the five-scenario set (representative weeks, hourly). Additions in MW/MWh, overnight capex in MEUR, expected total annualised cost and severe-drought cost in MEUR/yr, expected imports in TWh/yr; ND marks the non-dominated set in (expected cost, expected imports).

Strategy	+PV	+Wind	+B MW	+B MWh	Capex	E[cost]	E[imp.]	Cost sev.	ND
A. Baseline	0	0	0	0	0	109.4	2.27	320.4	
B. Solar expansion	629	0	0	0	500	70.9	1.64	283.5	
C. Wind expansion	0	418	0	0	500	112.1	2.07	323.7	
D. Battery expansion	0	0	522	2 087	500	164.9	2.83	373.4	
E. Coordinated (deterministic optimum)	629	0	0	0	500	70.9	1.64	283.5	
F. Climate-robust (stochastic, CVaR λ=1)	629	0	0	0	500	70.9	1.64	283.5	
Budget 100 MEUR (det.)	126	0	0	0	100	101.4	2.13	312.9	
Budget 250 MEUR (det.)	315	0	0	0	250	89.7	1.93	301.8	
Budget 750 MEUR (det.)	944	0	0	0	750	53.7	1.46	266.5	
Budget 1000 MEUR (det.)	1 258	0	0	0	1 000	37.9	1.37	251.2	
Budget 1500 MEUR (det.)	1 888	0	0	0	1 500	12.9	1.27	225.1	
Budget 2000 MEUR (det.)	2 000	0	0	0	1 589	10.4	1.26	222.3	•
Stochastic λ=0.0 (500 MEUR)	629	0	0	0	500	70.9	1.64	283.5	
Stochastic λ=0.25 (500 MEUR)	629	0	0	0	500	70.9	1.64	283.5	
Stochastic λ=4.0 (500 MEUR)	629	0	0	0	500	70.9	1.64	283.5	
Stochastic λ=0.0 (1500 MEUR)	1 888	0	0	0	1 500	12.9	1.27	225.1	
Stochastic λ=4.0 (1500 MEUR)	1 888	0	0	0	1 500	12.9	1.27	225.1	

7.9 Validity of the representative-week approximation

Seven plans were re-evaluated with full-year chronological hourly dispatch on all five scenarios. With the water-value boundary, the representative-week model's expected total cost differs from chronology by 1.7 to 10.0 MEUR/yr in absolute terms. It overstates the cost of the PV plans by 4.2–8.7 MEUR/yr and understates the wind plan by 10.0 MEUR/yr. Both models find no unserved energy for any plan, and the strategies rank in the same order (ties between identical plans aside). An earlier variant pinned each week's terminal volume ($V_T \geq \bar{V}^T$). That variant overstated cost

by up to 20.4 MEUR/yr and produced spurious shortages in the sampled drought weeks. Those shortages made a PV–battery portfolio look better on tail risk, but full chronology showed this to be an artefact: the PV-only plan had lower expected cost (66.7 vs 76.3 MEUR/yr) and lower CVaR (443.5 vs 455.7 MEUR/yr). We report this as a methodological finding. How the seasonal boundary of a representative period is treated can change the investment recommendation, and validation against full chronology should be routine.

Table 13: Full-year chronological (8760 h, hourly) versus representative-week (8 weeks, water-value boundaries) evaluation of the same plans on the five-scenario set. Costs in MEUR/yr; $\text{CVaR}_{0.9}$ of total annualised cost; ENS expected MWh/yr.

Plan	E[cost] chrono	E[cost] rep.	Diff.	CVaR chrono	CVaR rep.	ENS chrono
A. Baseline	107.7	109.4	1.7	525.8	524.0	0
B. Solar expansion	66.7	70.9	4.2	443.5	447.7	0
C. Wind expansion	122.1	112.1	-10.0	520.0	502.9	0
D. Battery expansion	162.8	164.9	2.1	577.9	576.9	0
F. Climate-robust (stochastic, CVaR λ=1)	66.7	70.9	4.2	443.5	447.7	0
Budget 1500 MEUR (det.)	4.2	12.9	8.7	308.6	321.4	0
Stochastic λ=0.0 (500 MEUR)	66.7	70.9	4.2	443.5	447.7	0

8 Sensitivity and robustness

Tornado analysis of the investment decision. For each of nine parameters we re-optimised the normal-year investment problem (500 MEUR, representative weeks) at a low and a high value (Fig. 11a). The optimal annualised system cost is most sensitive to the following parameters, in decreasing order of effect:

- drought severity: +213.5 MEUR/yr at $d = 0.3$;
- demand ($\pm$10%): $-74.3/+68.4$ MEUR/yr;
- the export/import price ratio: $+30.0/-18.8$ MEUR/yr at 0.75/0.95;
- PV capex ($\pm$30%): $-34.7/+19.1$ MEUR/yr;
- the import price ($\pm$30%): $+13.3/-13.3$ MEUR/yr;
- the discount rate (4%/10%): $-10.9/+12.1$ MEUR/yr.

The *technology choice*, all PV, is unchanged across every case. Only the PV amount responds, to its own capex: 899 MW at $\times$0.7 and 484 MW at $\times$1.3. The budget binds in both cases. VOLL and battery capex have no effect because no shortage occurs and no battery is chosen. Interconnector capacity between 600 and 1,200 MW moves cost by less than 3 MEUR/yr in a normal year. The importance of interconnection appears only under joint stress (Experiments F, G and J).

Drought severity sweep. Increasing the inflow reduction d from 0 to 0.6 in steps of 0.05, with the existing fleet, raises imports almost linearly: from 2.0 to 4.9 TWh/yr with full transfer availability, while annual cost rises from 41 to 481 MEUR (Fig. 6a). No shortage occurs at any severity with full availability. With 50% availability the system copes up to $d = 0.4$, and unserved energy then jumps to 341 GWh at $d = 0.5$ and 956 GWh at $d = 0.6$ (Fig. 6b). The reliability boundary of the system therefore lies in the joint space of drought severity and import availability.

Battery value across drought and import stress. With import availability at 60%, the net benefit of a 4-hour battery (operating saving minus annualised cost) is negative up to drought severity $d = 0.4$: between -10 and -97 MEUR/yr for 400–3,200 MWh (Fig. 11b). It turns strongly positive at $d = 0.6$, where shortage occurs: +248 to +332 MEUR/yr. There the battery

reduces unserved energy from 281 to between 228 and 192 GWh, and each avoided MWh is valued at VOLL. Batteries in this system are therefore *reliability* assets under joint drought and import stress, not arbitrage assets. Their economic case depends on the probability of such joint events and on the VOLL assumed for them. This is the case where the VOLL assumption matters most.

Temporal resolution. The operating saving from a 400 MW / 1,600 MWh battery is 9.8, 8.8, 7.0 and 1.3 MEUR/yr at 1-, 2-, 3- and 6-hour time steps, and its cycling falls from 401 to 111 equivalent full cycles (Fig. 12). Six-hourly steps remove 87% of the battery's arbitrage value. This explains why the first, 6-hourly version of our stochastic model selected no storage, and why the investment studies use hourly representative weeks.

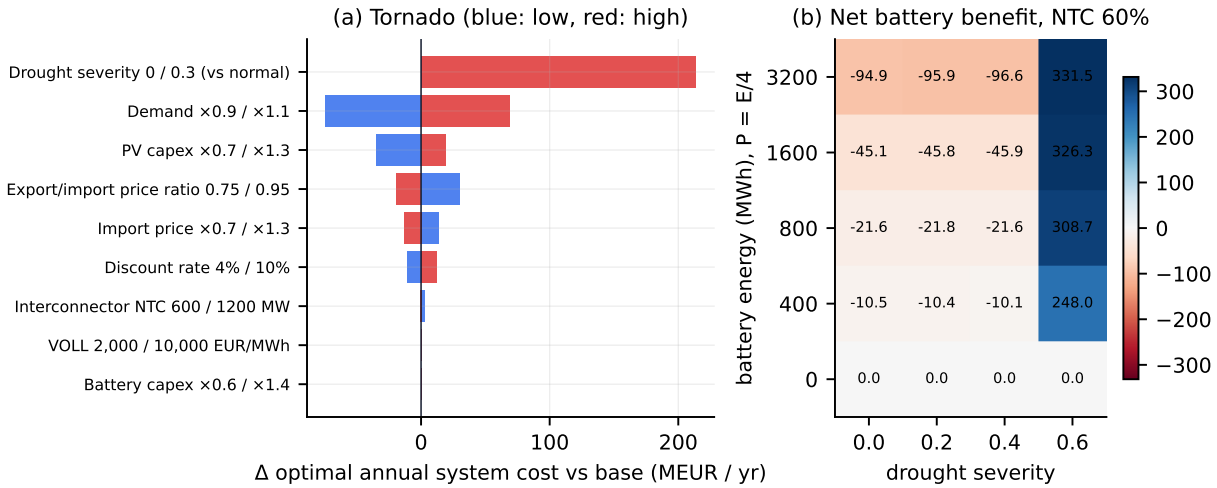


Figure 11: (a) Change in the optimal annualised system cost when each parameter is set to its low (blue) or high (red) value, with investment re-optimised each time. (b) Net annual benefit (MEUR/yr) of a 4-hour battery as a function of drought severity and battery size, with import availability 60%.

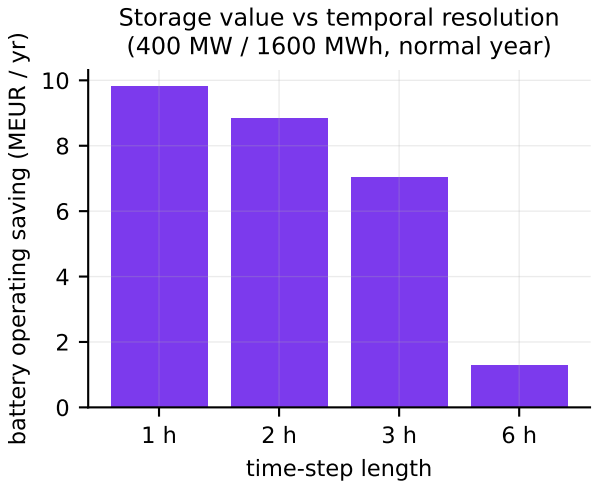


Figure 12: Operating value of a 400 MW / 1,600 MWh battery in the normal year as a function of the time-step length.

9 Discussion

Drought as financial risk, interconnection as reliability risk. The clearest result is a separation of risks. With the interconnector available, even a 40% inflow reduction is met entirely

by imports. The cost of drought is financial: operating cost rises by 284 MEUR/yr between the normal and the severe-drought year. Unserved energy appears only when transfer availability falls to about half, and it is energy-limited. Short-duration storage can therefore reduce it only partially (13% at 50% availability). At an assumed VOLL of 5,000 EUR/MWh, however, even that partial reduction is worth more than the battery costs. Batteries thus earn their keep here as reliability assets in joint stress events, not as arbitrageurs. This agrees with the concern of van Vliet et al. [5] that climate impacts on water-dependent generation are partly compensated by trade and partly transmitted to neighbours. It also suggests that, for Albania, the joint probability of domestic drought and reduced regional export capacity deserves more attention than domestic drought alone. Because Balkan hydro systems share weather, this joint risk is plausible. Our scenario probabilities for it are assumed, not estimated.

Solar as a no-regret option, conditional on export. Under the base assumptions, PV is worth about 1.7× its annualised cost at the margin. It dominates all other investments at every budget and in every scenario, which is why VSS is zero. Two mechanisms drive this. First, in a reservoir system PV saves water, which the reservoir then moves to high-price hours, so PV inherits part of the reservoir's flexibility value. Second, surplus PV can be exported at 85% of the import price. The decline of renewable market value with penetration described by Hirth [20] is therefore weak here, but only because the export route is assumed to absorb the surplus. When exports are restricted (Experiment I2), this conclusion changes. That restriction is the condition under which planning for uncertainty starts to matter.

Storage: timing, not volume. Batteries reduce cost by shifting energy in time. In doing so they *increase* gross imports, by importing cheaply and serving the evening peak. Policy goals stated as "reducing import dependence" and "reducing system cost" can therefore point in different directions for storage. Arbitrage alone does not recover battery costs here: savings are 13–27% of annualised cost across the tested sizes and years. The economic case for batteries rests instead on avoided shortage in joint drought and import-stress events (Section 8), and so on how likely those events are and what VOLL is assumed. The marginal analysis shows, however, that discharge power is worth more than its cost once some energy capacity exists. This points to shorter-duration systems and to the reserve and ancillary services that the model omits [15, 16]. The reservoir already supplies long-duration flexibility: additional reservoir *volume* has zero marginal value in every case, because annual inflow, not storage size, is the binding resource.

Methodological lessons. Two modelling choices changed answers by more than many of the parameter uncertainties. First, temporal resolution: at 6-hourly steps battery value nearly vanishes and stochastic planning had nothing to choose between (Section 8), consistent with Poncelet et al. [17]. Second, how the boundaries of representative periods are handled for a seasonal store. Pinning each week's terminal volume created spurious scarcity. That made a battery look valuable for tail risk and reversed the ranking of two plans compared with full chronology. Valuing the end volume at the coarse annual run's water value, a single-cut future-cost function as in SDDP [6], removed the artefact. This is a concrete example of the seasonal-storage problem in representative-period models described by Kotzur et al. [18]. Our recommendation is that studies of hydro-dominated systems validate their period selection against full chronology, as we do here.

Model implications versus policy. These are implications of a single-node model with synthetic demand and prices. They are not recommendations for Albanian policy. A credible policy analysis would need observed hourly load (ENTSO-E), regional price and transfer-capacity data, calibrated hydrology and a network representation. The open pipeline is designed so that each of these can be substituted without code changes to the optimisation core.

10 Limitations

Network representation. The model has a single node. Internal transmission constraints, losses and stability limits are absent. The four interconnections (Montenegro, Kosovo, North Macedonia, Greece) are collapsed into one import and one export limit, so this is not a power-flow model. Regional correlation of droughts and prices is represented only through the scenario availability factor a_s.

Market data. Demand and prices are synthetic. Export revenue is a fixed 85% of the import price. Many conclusions, especially the value of PV, which in this model can always be exported up to 900 MW, depend on this assumption. With observed ENTSO-E load and regional day-ahead prices the analysis could be repeated without code changes.

Hydrology. Inflows are reconstructed from reanalysis precipitation with an uncalibrated conceptual model and scaled to an assumed mean energy. They are not gauged inflows. Drought is a uniform scaling of one reference year. Real droughts also change the seasonal timing of inflows, and this is not captured.

Reservoir simplification. The three-stage Drin cascade is one equivalent reservoir with a fixed aggregate head. Head-dependent efficiency, travel times, individual reservoir limits and flood-control rules are omitted. The other hydro plants are treated as run-of-river without storage.

Operational detail. There is no unit commitment, no ramping, no reserve or ancillary-service requirement and no thermal fleet. Batteries are therefore valued only for energy arbitrage and shortage avoidance, which probably understates their real value. Simultaneous charging and discharging is excluded by economics, not by binaries, and is monitored.

Time structure. Investment and stochastic studies use eight representative weeks with fleet-independent seasonal boundaries. Water allocation across seasons therefore does not adapt to new investment. We quantify the error against full-year chronology (Section 7.9), but the planning results inherit it.

Costs and uncertainty. Technology costs are global IRENA averages with an assumed regional multiplier. VOLL, the discount rate, lifetimes and the scenario probabilities are assumptions. Five scenarios cannot describe the full climate distribution, and the 15% stress probability is a judgement, not an estimate.

Validation. The model has been *verified*: against an independent PyPSA formulation and through automated conservation and behaviour tests. It has **not been validated** against observed operations of the Albanian system. Scenario comparisons are model experiments, not empirical evidence. The results are model implications under stated assumptions and should not be read as policy recommendations.

11 Conclusion

We built and verified an open optimisation model of a hydropower-dependent electricity system. It couples hourly dispatch, a physical reservoir water balance, batteries, imports and investment under uncertainty, and it reports every result together with its data provenance and its limitations. Under the stated (partly synthetic) assumptions, the experiments support the following defensible findings.

1. Drought is primarily a financial and import-dependence risk. Reliability is governed by the availability of interconnection: shortages require import availability of about 50% or less together with severe drought or demand growth.
2. Solar PV is a no-regret investment in every scenario as long as its surplus can be exported. Its marginal value is about 1.7× its annualised cost, and the technology choice is robust to every sensitivity tested.
3. Batteries do not recover their cost through arbitrage, and they increase gross imports. They do pay off through avoided unserved energy in joint drought and import-stress events. Discharge power is more valuable than additional energy capacity.
4. Stochastic, CVaR-based planning changes the decision only when the dominant resource saturates. With exports restricted, it buys a wind hedge that lowers tail cost by 74.5 MEUR/yr for 15.1 MEUR/yr of expected cost, and this is confirmed by full chronology.
5. Methodologically, temporal resolution and the boundary treatment of representative periods for seasonal storage each changed conclusions. Water-value boundaries and chronological validation are inexpensive safeguards.

Next steps. (i) Replace the synthetic demand and prices with observed ENTSO-E load and regional market data. (ii) Calibrate the hydrology against gauge or plant records and represent the three Drin stages separately, with variable head. (iii) Add network constraints for the four interconnections and correlated regional scenarios, including the low-availability regime in which storage becomes valuable. (iv) Add reserve requirements and multi-stage hydro scheduling, for example SDDP-style water-value cuts. (v) Estimate scenario probabilities from climate-model ensembles rather than by assumption.

Data and code availability

All code, input-data provenance records, experiment outputs and the scripts that generate every figure and table in this manuscript are contained in the HYDROFLEX AI repository. Running `python -m hydroflex.experiments.run_all` regenerates the experiment files, tables and figures. Demand and price series are synthetic and are labelled as such in every output.

Use of software and AI assistance

Model code, the experiment pipeline and draft text were developed with the assistance of an AI coding assistant (Claude, Anthropic) operating under the author's direction. All reported numbers are produced by executed code in the repository.

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A Symbols

Table 14: Symbols and units.

Symbol	Unit	Meaning
$t,\ \Delta t,\ T$	h	time step index, step length, number of steps
$s,\ \pi_s$	–	scenario and probability
x_{pv}, x_w	MW	installed PV and wind capacity (first stage)
x_{bp}, x_{be}	MW, MWh	battery power and energy capacity (first stage)
$x_{imp},\ x_{vmax}$	MW, hm^3	import capacity, maximum reservoir volume (fixed; marginal-value analysis)
$x_k^0,\ \bar{x}_k$		existing capacity, maximum addition
$A_k,\ C_k$	kEUR/unit(-yr)	annualised and overnight cost
w	–	$T\Delta t/8760$ horizon weight of annual costs
g^{pv}, g^{w}, g^{ror}	MW	PV, wind, run-of-river generation
f_t^{pv}, f_t^{w}	p.u.	capacity factors
$R_{s,t}$	MW	run-of-river availability
h	MW	reservoir hydro output
$V,\ I,\ \sigma$	hm^3, hm^3/h	reservoir volume, inflow, spill
κ	$\mathrm{MWh/hm}^3$	energy equivalent of water, $\rho_w g H \eta \cdot 10^6/3.6\times10^9$
$P^h,\ r^{\min}$	MW, hm^3/h	turbine capacity, minimum release
$c,\ d,\ e$	MW, MW, MWh	battery charge, discharge, state of charge
$\eta^c, \eta^d, \varepsilon, s_0, \delta^{\max}$	–	efficiencies, minimum SOC fraction, initial SOC fraction, maximum duration (h)
$m,\ y,\ a_s,\ Y$	MW	imports, exports, import availability factor, export limit
$p^m,\ \rho$	EUR/MWh, –	import price, export/import price ratio
u, VOLL	MW, EUR/MWh	unserved energy, value of lost load
L_s	kEUR	scenario operating cost
$\alpha,\ \lambda,\ \zeta,\ \phi_s$	–	CVaR level, risk weight, VaR variable, tail excess
B	MEUR	overnight capital budget

B Additional results

Table 15: Experiment D: battery storage without new renewables (annual, hourly).

Case	Imports (TWh)	Net imp. (TWh)	Curtail. (GWh)	Batt. cycles	ENS (MWh)	Opex (MEUR/yr)
Normal, 0 MW / 0 MWh	2.00	0.77	0.0	0.0	0	40.8
Normal, 200 MW / 800 MWh	2.23	0.81	0.0	443.0	0	35.3
Normal, 400 MW / 1600 MWh	2.42	0.84	0.0	401.4	0	31.0
Normal, 800 MW / 3200 MWh	2.60	0.86	0.0	280.0	0	27.0
Severe, 0 MW / 0 MWh	3.79	3.48	0.0	0.0	0	324.5
Severe, 200 MW / 800 MWh	4.01	3.52	0.0	510.9	0	317.5
Severe, 400 MW / 1600 MWh	4.20	3.56	0.0	463.8	0	311.8
Severe, 800 MW / 3200 MWh	4.38	3.59	0.0	322.6	0	306.5

Table 16: Experiment G: demand growth stress test under severe drought (annual, hourly).

Case (severe drought)	Imports (TWh)	LOLH (h)	ENS (MWh)	Peak short. (MW)	Cost (MEUR/yr)
growth 0%, NTC 100%	3.79	0	0	0	324.5
growth 10%, NTC 100%	4.47	0	0	0	408.7
growth 20%, NTC 100%	5.17	0	0	0	495.2
growth 30%, NTC 100%	5.91	0	0	0	585.5
growth 0%, NTC 50%	3.49	0	0	0	356.8
growth 10%, NTC 50%	3.82	1 465	455 643	840	2 683.3
growth 20%, NTC 50%	3.88	3 061	1 201 086	1 364	6 415.7
growth 30%, NTC 50%	3.91	4 088	1 969 392	1 523	10 260.2

Table 17: Risk-weight sweep of the two-stage programme ($\alpha = 0.9$). Costs in MEUR/yr; time is wall-clock for both boundary passes.

λ	Budget	+PV	+Wind	+B MW	+B MWh	E[cost]	$\text{CVaR}_{0.9}$	Time (s)
0.00	500	629	0	0	0	70.9	397.3	23.1
0.25	500	629	0	0	0	70.9	397.3	21.2
1.00	500	629	0	0	0	70.9	397.3	23.1
4.00	500	629	0	0	0	70.9	397.3	28.0
0.00	1 500	1 888	0	0	0	12.9	170.2	15.3
4.00	1 500	1 888	0	0	0	12.9	170.2	20.7

C Solver diagnostics and verification

Table 18: Solver diagnostics (HiGHS via highspy) for representative problems.

Problem	Method	Vars	Rows	Non-zeros	Solve (s)	Simplex it.	IPM it.	Status
Annual dispatch, hourly	choose	105 128	105 126	306 611	18.10	28 525	0	Optimal
Stochastic, 8 weeks × 5 scen.	simplex	80 732	80 847	269 256	14.10	41 016	0	Optimal
Investment, 8 weeks, normal	simplex	16 150	16 170	47 126	0.87	8 275	0	Optimal

Table 19: Independent verification of the native sparse LP against a PyPSA/linopy formulation of the same system (both solved with HiGHS). Times in seconds include model construction. Reservoir trajectories may differ where the LP optimum is not unique.

Scenario	Hours	Invest	Native (kEUR)	PyPSA (kEUR)	Rel. diff.	max$\|\Delta V\|$	t native	t PyPSA
normal	168	no	-4 170.98	-4 170.98	0.0e+00	0.00e+00	0.05	28.60
severe drought	168	no	1 980.89	1 980.89	0.0e+00	0.00e+00	0.05	4.45
compound shock	168	yes	-6 408.66	-6 408.66	0.0e+00	2.21e-01	0.11	4.76
normal	720	no	-8 057.99	-8 057.99	0.0e+00	2.72e+00	0.21	4.49
severe drought	720	yes	-3 637.15	-3 637.15	0.0e+00	5.81e-01	0.43	5.74

D Data dictionary

Table 20: Data dictionary: machine-readable provenance records in `data/provenance/`.

Dataset id	Provider	Resolution	Coverage	Nature
nasa_power_daily_drin_black	NASA Langley Research Center, POWER Project	daily	2000-01-01 to 2023-12-31	observed-derived reanalysis precipitation; downstream inflows are reconstructed, NOT gauged
nasa_power_daily_drin_koman	NASA Langley Research Center, POWER Project	daily	2000-01-01 to 2023-12-31	observed-derived reanalysis precipitation; downstream inflows are reconstructed, NOT gauged
nasa_power_daily_drin_kukes	NASA Langley Research Center, POWER Project	daily	2000-01-01 to 2023-12-31	observed-derived reanalysis precipitation; downstream inflows are reconstructed, NOT gauged
nasa_power_daily_drin_white	NASA Langley Research Center, POWER Project	daily	2000-01-01 to 2023-12-31	observed-derived reanalysis precipitation; downstream inflows are reconstructed, NOT gauged
nasa_power_hourly_load_tirana	NASA Langley Research Center, POWER Project	hourly	2019-01-01 to 2023-12-31	observed-derived reanalysis / satellite estimate (NOT measured power production)
nasa_power_hourly_solar_fier	NASA Langley Research Center, POWER Project	hourly	2019-01-01 to 2023-12-31	observed-derived reanalysis / satellite estimate (NOT measured power production)
nasa_power_hourly_solar_vlore	NASA Langley Research Center, POWER Project	hourly	2019-01-01 to 2023-12-31	observed-derived reanalysis / satellite estimate (NOT measured power production)
nasa_power_hourly_wind_lezhe	NASA Langley Research Center, POWER Project	hourly	2019-01-01 to 2023-12-31	observed-derived reanalysis / satellite estimate (NOT measured power production)
nasa_power_hourly_wind_vlore	NASA Langley Research Center, POWER Project	hourly	2019-01-01 to 2023-12-31	observed-derived reanalysis / satellite estimate (NOT measured power production)
natural_earth_admin0_50m	Natural Earth (naturalearthdata.com)	n/a	static	observed (cartographic)
open_meteo_era5_tirana_2021	Open-Meteo.com (Copernicus ERA5 reanalysis)	hourly	2021-01-01 to 2021-12-31	reanalysis
processed_hourly_profiles	Derived by HydroFlex AI from NASA POWER (see raw provenance records)	hourly	2019–2023, 8760 h per year (29 Feb removed in leap years)	mixed (see record)
processed_inflow_daily	Derived by HydroFlex AI from NASA POWER daily precipitation and temperature	daily	2001-01-01 to 2023-12-31 (2000 used as spin-up)	reconstructed — volume scaled to an assumed long-run mean energy; NOT observed inflows

E Reproducibility instructions

1. `python -m venv .venv` and `pip install -e ".[dev]"` (Python $\geq$ 3.11).
2. Optional data refresh:
   `python -m hydroflex.data.fetch`, then
   `python -m hydroflex.data.profiles`.
3. `python -m hydroflex.experiments.run_all` (add `-force` to recompute cached experiments).
4. `pytest` runs the verification suite.

5. Compile this manuscript with `latexmk -pdf main.tex` (or `tectonic main.tex`) in `research/paper`.

Software versions, seeds and stage timings are recorded in `results/experiments/manifest.json`.